A Novel Weld Defect Detection Method for Intelligent Magnetic Flux Leakage Detection System via Contextual Relation Network

Xiangkai Shen, *Graduate Student Member, IEEE*, Jinhai Liu, *Senior Member, IEEE*, Lin Jiang, *Member, IEEE*, Xiaoyuan Liu, *Student Member, IEEE*, and Huaguang Zhang, *Fellow, IEEE*

***Abstract*—Accurate weld defect detection is a critical signal processing technique for intelligent magnetic flux leakage (MFL) detection system. However, the MFL signals between the defect and the weld are coupled with each other, and the features between the two are difficult to distinguish, resulting in unsatisfactory accuracy of weld defect detection. To address this issue, a novel weld defect detection method named contextual relation network (CRNet) is proposed to achieve high-accuracy weld defect detection. First, based on the strong coupling relation of MFL signals, a feature extraction network with sparse attention is proposed to extract high-quality features in a sparse manner. Second, a multiscale relation exploration module is proposed to adaptively aggregate multiscale features and fully mine the coupling relation between defect and weld, so that the contextual dependencies between them can be distinguished, and more valid semantic features can be acquired. Third, a multiview collaborative detection mechanism is proposed to realize complementary and coarse-to-fine decisions, so that more accurate detection results can be obtained. The CRNet can clarify the coupling relation and feature differences between defect and weld, guaranteeing independent feature extraction and high detection accuracy. Experiments show that CRNet can reach an average detection accuracy of 95.5%, which is more effective than advanced methods and has strong practicality.**

***Index Terms*—Contextual relation network (CRNet), magnetic flux leakage (MFL), multiview collaborative mechanism, weld defect detection.**

Manuscript received 16 January 2023; revised 5 May 2023; accepted 2 July 2023. Date of publication 26 July 2023; date of current version 19 January 2024. This work was supported in part by the National Natural Science Foundation of China under Grant U21A20481 and Grant 61973071, in part by the LiaoNing Revitalization Talents Program under Grant XLYC2002046, and in part by the Fundamental Research Funds for the Central Universities of China under Grant N2104020. *(Corresponding author: Jinhai Liu.)*

Xiangkai Shen, Lin Jiang, and Xiaoyuan Liu are with the College of Information Science and Engineering, Northeastern University, Shenyang 110819, China (e-mail: 2010302@stu.neu.edu.cn; 1810278@stu.neu.edu.cn; 2010301@stu.neu.edu.cn).

Jinhai Liu and Huaguang Zhang are with the State Key Laboratory of Synthetical Automation for Processes Industries, Shenyang 110819, China, and also with the College of Information Science and Engineering, Northeastern University, Shenyang 110819, China (e-mail: liujinhai@mail.neu.edu.cn; hgzhang@ieee.org).

Color versions of one or more figures in this article are available at https://doi.org/10.1109/TIE.2023.3294578.

Digital Object Identifier 10.1109/TIE.2023.3294578

I. Introduction

THE integrity of pipelines is an important guarantee for the safe transportation of oil and gas resources [1], [2]. Due to imperfections in welding technology, corrosion of the transmission medium, and lack of a protective layer, etc., pipelines often produce weld defects [3], [4]. Weld defects are always the weak point of the pipeline. Compared with other corrosion defects, weld defect arises in a more sensitive location, which is more likely to result in pipeline rupture and is fatal to safe pipeline operation. If weld defect is not detected accurately, the risk of pipeline leakage will increase, leading to serious threats to the surrounding environment and people. Therefore, accurate weld defect detection is an urgent and necessary task. Magnetic flux leakage (MFL) testing is widely used for the online detection of weld defect in long-distance complex transport pipelines, which are offshore, buried, or inaccessible [5], [6]. The MFL signals from the inspection robot are automatically processed for weld defect detection [7]. However, the MFL signals between defect and weld are coupled to each other, which makes it hard to distinguish between the features of defect and weld, so that weld defect is easily missed or misdiagnosed. In particular, if weld defect is missed, this increases the risk of pipeline failure and, thus, fails to ensure the safe transportation of oil and gas resources. Therefore, weld defect detection is a challenging and important issue in the MFL signal processing field. All the time, various studies dedicate to weld defect detection, which have profound theoretical and engineering implications.

Early weld defect detection methods often focus on signal conversion [8], [9] and feature selection techniques [10], [11]. The discrete cosine transform [8] and magneto-optical imaging [9] were proposed to detect weld defect in pipelines. In [10] and [11], spatial–temporal regularization (STLR) and principal component analysis were proposed to realize weld defect detection. Although the simple operation of the above methods can achieve excellent performance, they are sensitive to artificially selected signal processing methods and manual features. Due to the nonlinear and coupling relation between defect and weld, it is difficult to determine effective signal conversions and feature selection methods. Hence, the accuracy and generalization of the above methods are degraded.

0278-0046 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 17,2026 at 03:48:22 UTC from IEEE Xplore. Restrictions apply.

Recently, deep learning (DL) methods have made breakthroughs in various industrial inspection fields, such as fault diagnosis, remaining life prediction, etc. In [12], a circular teacher network was designed to achieve accurate pipeline fault diagnosis with limited labeled samples. In [13], a deep multisource parallel bilinear fusion network was proposed to effectively realize the prediction of the milling cutters' remaining life. Based on the above excellent performance, the DL methods are considered to be a promising and reasonable way to detect weld defects. The DL methods for weld defect detection can be divided into two main frameworks. One is the region-level detection framework [14], [15], [16], where the detection results are usually given by bounding boxes. However, they can only provide a rough area of defect and cannot precisely describe defect boundaries, especially for defects with complex and irregular shapes. The other current acceptable one is the pixel-level detection framework [17], [18], [19], [20], compared with the former, where defects' boundaries rather than the bounding boxes can be obtained via pixelwise segmentation, so that higher detection accuracy can be provided. In [21] and [22], deep belief network and fully convolutional network (FCN) were applied to realize weld defect detection and crack detection. In [23], an attention-guided segmentation network was proposed to achieve pixel-level weld defect detection. In [24], the DeepLab with multiscale attention was proposed to detect weld crack defects. Despite the impressive performance of the above methods, considering the strong coupling between defect and weld, existing methods are difficult to apply directly to MFL weld defect detection. Specifically, MFL weld defect detection still faces two dilemmas: 1) How to reduce the low-quality weld features coupled in the defect features and guarantee effective feature extraction. 2) For coupled region signals, how to effectively explore the coupling relation between defect and weld, so as to clarify the differences between interclass features and obtain accurate detection results.

To overcome the first dilemma, based on the strong coupling relation of MFL signals, a feature extraction network with sparse attention is proposed to filter low-correlation features and maximize the high-quality features, which ensures effective feature extraction. The proposed feature extraction network includes a pretrained encoder module and a sparse attention module (SAM). The pretrained model is obtained by training on large public datasets, which guarantees the validity of shallow feature extraction by fine-tuning a few parameters. In addition, the proposed SAM can select highly correlated defective features in a sparse manner and reduce the low-quality weld features coupled in the defect features, so that more effective deep features can be obtained.

To overcome the second dilemma, a multiscale relation exploration (MRE) module and a multiview collaborative detection (MCD) mechanism are proposed to mine the coupling relation between defect and weld and obtain high-accuracy detection results. Specifically, MRE module specifies the contextual dependencies between defect and weld based on a learnable relation matrix, which guarantees intraclass feature aggregation and clarifies interclass feature differences. In addition, MCD mechanism makes the network learn the feature representation of coupled regions from multiple levels. It realizes the complementary and coarse-to-fine decision-making judgment of coupled regions, so that accurate detection results can be obtained.

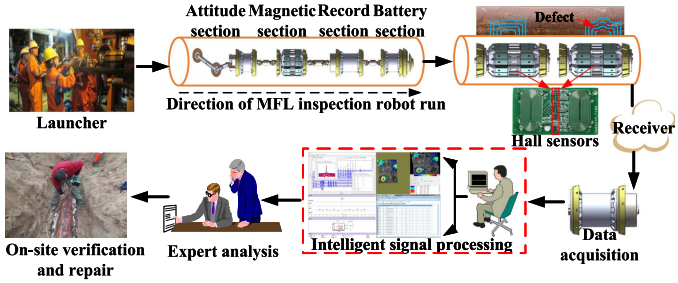


Fig. 1. Overall industrial process of the MFL detection system.

Regarding the above analysis, a novel weld defect detection method named contextual relation network (CRNet) is proposed to realize high-accuracy weld defect detection. In general, the proposed method guarantees independent and effective feature extraction. In addition, the coupling relation between defect and weld is fully explored, and the interclass features in the coupling region can be effectively distinguished. To the best of our knowledge, this article is the first to explore the coupling relation between MFL weld defect and weld and to complete a highly accurate pixel-level weld defect detection.

This article provides the following contributions.

1) A feature extraction network with sparse attention is proposed, which reduces the strong correlation between defect and weld in a sparse manner, so that the weld features coupled in the defect features can be reduced and the high-quality defect features can be fully extracted.
2) An MRE module is proposed to mine the coupling relation between defect and weld. The proposed module explores the contextual dependencies between defect and weld and specify the differences in interclass features of the coupling region, so that the accuracy of weld defect detection can be improved.
3) An MCD mechanism is proposed to learn the feature representation of coupled regions from multiple levels, which realizes complementary and coarse-to-fine decision-making judgments, so that more accurate weld defect detection results can be obtained.

The rest of the article is organized as follows. Section II introduces the MFL detection system and the problem to be solved. Section III describes the proposed method in detail. Experimental details are stressed in Section IV. Finally, Section V concludes this article.

II. MFL Detection System and Problem Statement

A. MFL Detection System

The MFL detection system is shown in Fig. 1, when carrying out pipeline inspection works, the MFL inspection robot is sent to the pipeline from the "Launcher" and recycled from the "Receiver." In this process, the pipeline wall is magnetized and a saturated magnetic field is generated. When the pipeline is undamaged, magnetic field lines are evenly distributed. If there

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 17,2026 at 03:48:22 UTC from IEEE Xplore. Restrictions apply.

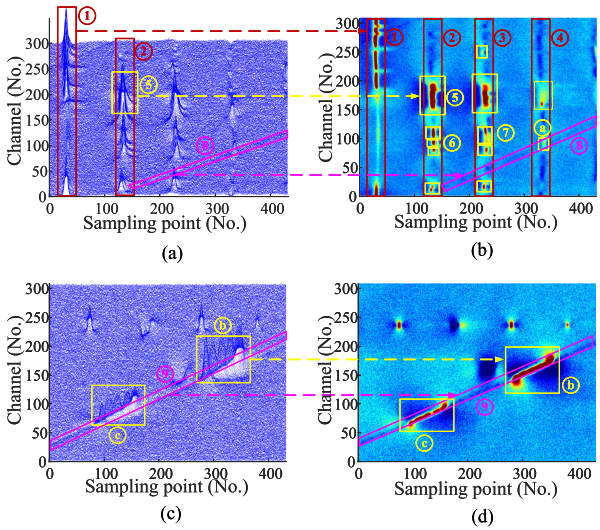


Fig. 2. Examples of weld defects. (a) and (b) Signals and 2-D displays of girth weld defects. (c) and (d) Signals and 2-D displays of spiral weld defects.

are defects on the pipeline, the magnetic field lines leak from the defect location into the air and are picked up by the running MFL inspection robot. At the end of the inspection, the MFL inspection robot completes the acquisition of MFL signals, where signals are the amplitude of the leakage flow measured by Hall sensors. Researchers and experts analyze MFL signals offline to diagnose the safety status of pipelines. Based on the analysis results, engineers repair damaged pipelines in a timely manner.

B. Problem Statement

The core of the research in this article is to solve the problem of low accuracy of weld defect detection. Weld defect is a serious threat to the safe operation of the pipeline. The strong coupling relation between defects and welds poses a challenge to the high-accuracy detection of weld defects. To describe the challenge clearly, weld defects have been highlighted in Fig. 2, where Fig. 2(a) and (c) shows signal curves and Fig. 2(b) and (d) shoes 2-D displays. As shown in Fig. 2, defects (⑤⑥⑦) and defects (ⓐⓑⓒ) are located in the girth welds (②③) and spiral welds (⑧⑨), respectively. The high-coupling relation between the defect signal and the weld signal makes the features of the two mixed, which makes the detection accuracy unsatisfactory.

To address the above problems, we hope to explore the coupling relation between defect and weld. Specifically, inspired by previous studies, the problem is converted to a supervised semantic segmentation model-building problem. Therefore, a novel semantic segmentation method named CRNet is proposed. Define the CRNet and model building process as $\mathbb{N}$ and $\mathbb{B}$. Given the weld defect dataset and ground truth as $\mathbb{D} = \{d_i\}_{i=1}^{N}$ and $\mathbb{G} = \{g_i\}_{i=1}^{N}$. The model-building process is stated as an optimization problem in

$$\mathbb{B} = \arg\min_{\delta} \sum_{i=1}^{N} \mathcal{L}(\mathbb{N}(d_i, \delta), g_i) \tag{1}$$

where N denotes the number of samples. δ and $\mathcal{L}$ denote parameters of CRNet and loss function, respectively. The built model will be applied to weld defect detection. The details of the CRNet are introduced in Section III.

III. Framework of the Proposed Method

A. Overview of the Proposed Method

As illustrated in Fig. 3, the proposed method consists of four phases. In the *training samples standardization* phase, original MFL signals are unified and transformed into refined samples. In the *feature extraction* phase, the effective sparse defect features are fully extracted from the refined samples. In the MRE phase, multiscale sparse features are adaptively fused. In addition, the designed learnable relation matrix explores the coupling relation between defect and weld, so that segmentation features are generated. In the MCD phase, a multiview collaborative mechanism is proposed to realize weld defect detection based on segmentation features.

B. Training Samples Standardization

The samples standardization is designed to reduce the influence of negative factors and obtain refined samples, which includes signal unification and signal transformation.

1) Signal Unification $c(\cdot)$***:*** The signal unification includes filtering, baseline correction, and interpolation. Gaussian filtering and a cubic spline interpolation are applied to ensure the validity and continuity of the signal. In addition, considering the difference in the base value of the Hall sensor, the MFL signals are processed by baseline correction in

$$H_{i,j} = X_{i,j} - \frac{1}{q}\sum_{m=1}^{q} X_{m,j} + \frac{1}{qe}\sum_{m=1}^{q}\sum_{n=1}^{e} X_{m,n} \tag{2}$$

where $X_{i,j}$ and $H_{i,j}$ are the original MFL signals and unified MFL signals, respectively. q and e are sampling points and the number of Hall sensors, respectively.

2) Signal Transformation $t(\cdot)$***:*** To highlight critical features, H (2-D) is converted to three-channel image D in

$$\begin{cases} D_{i,j} = C(s) \\ s = \left\lfloor \frac{H_{i,j} - \min(H)}{\max(H) - \min(H)} \times \xi \right\rfloor \end{cases} \tag{3}$$

where $D_{i,j}$ is the pixel values at position (i, j), C is color map, and ξ is the conversion resolution. The advantages of samples standardization are listed.

1) The signal unification eliminates the negative impact caused by sensor differences and noise during signal acquisition blue process, so that the validity of subsequent data analysis can be ensured.
2) Signal transformation highlights the critical features of defect and weld, so that the effectiveness of subsequent feature extraction can be guaranteed.

C. Feature Extraction

Due to the strong coupling of weld signals and defect signals, low-quality weld features mixed in defect features cannot be

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 17,2026 at 03:48:22 UTC from IEEE Xplore. Restrictions apply.

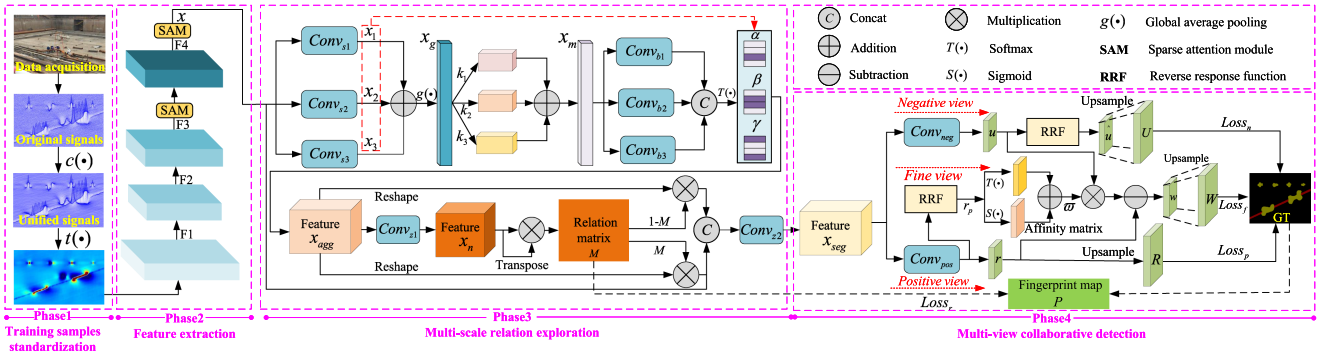


Fig. 3. Overview of the proposed CRNet.

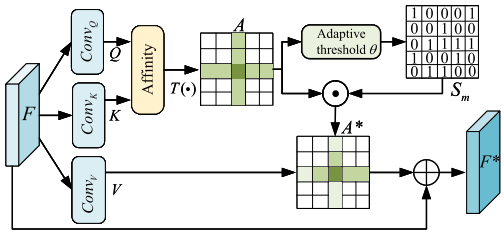


Fig. 4. Proposed SAM.

filtered. Thus, a feature extraction network with sparse attention is proposed to reduce low correlation features in a sparse manner, which includes encoder module $\mathbb{F}_e(\cdot)$ and SAM $\mathbb{F}_s(\cdot)$.

1) Encoder Module: The encoder module is built on pretrained ResNet-50, which contains four blocks. To ensure the extraction of shallow features and prevent overfitting, the parameters of the first two blocks are frozen, and remaining parameters are fine-tuned. The encoded feature f of the MFL signals is given $f = \mathbb{F}_e(D, \epsilon)$, where ϵ is the parameter of $\mathbb{F}_e$.

2) Sparse Attention Module: Self-attention mechanism is effective in capturing long-distance information and feature enhancement [25], [26], [27]. However, all features are involved in the aggregation of enhanced features. This means that low-correlation weld features will be coupled with the defect features, which reduces the defect detection accuracy. Therefore, an SAM is proposed, which can obtain long-range defect information in a sparse manner, so that the coupling features are reduced, and high-quality defect features are obtained.

As indicated in Fig. 4, given a feature map F, three 1×1 convolutions are applied to F to generate features Q, K, and V, where $\{Q, K\} \in \mathbb{R}^{\frac{C}{4} \times H \times W}$ and $V \in \mathbb{R}^{C \times H \times W}$. Inspired by Huang et al. [18], at each position l in the spatial dimension of Q, the correlation between elements in the same row and column as l is computed instead of all elements. Define $Q_l \in \mathbb{R}^{1 \times \frac{C}{4}}$, and $\Gamma_l \in \mathbb{R}^{(H+W-1) \times \frac{C}{4}}$ from the feature map K. Then, the affinity matrix $A \in \mathbb{R}^{(H \times W) \times (H+W-1)}$ is calculated in

$$A = T\left(\zeta(Q) \otimes {\Gamma_l}^{\top}\right) = \frac{\exp\left(\zeta(Q) \otimes ({\Gamma_l}^{\top})_i\right)}{\sum_{i=1}^{H+W-1} \exp\left(\zeta(Q) \otimes ({\Gamma_l}^{\top})_i\right)} \tag{4}$$

where $T(\cdot)$ and $\zeta(\cdot)$ denote Softmax and reshape, respectively. To further weaken the coupling feature, elements with higher correlation in A should be retained, and vice versa. Thereby, the sparse matrix $S_m \in \mathbb{R}^{(H \times W) \times (H+W-1)}$ is obtained in (5). Noteworthy, adaptive threshold θ is generated by 1-D convolution

$$S_m(i,j) = \begin{cases} 1, & A(i,j) \geq \theta \\ 0, & A(i,j) < \theta. \end{cases} \tag{5}$$

Then, the updated affinity matrix $A^* = S_m \bigodot A$, where $\bigodot$ denotes the Hadamard product. Finally, the enhanced features $F^* \in \mathbb{R}^{C \times H \times W}$ is calculated in

$$F^* = \mathbb{C}_{l=1}^{H \times W} \left(\sum_{i=0}^{H+W-1} A^*_{i,l} V_{i,l} + F_l \right) \tag{6}$$

where $V_{i,l}$ denotes the i_{th} adjacent feature vector at position l in the spatial dimension of V, and $\mathbb{C}(\cdot)$ denotes concatenation.

The advantages of the proposed feature extraction are listed as follows.

1) The pretrained encoder module ensures the validity of shallow feature extraction with reduced training parameters.
2) The proposed SAM reduces the coupling of defect and weld at the feature extraction level in a sparse manner, so that more high-quality defect sparse features can be acquired.

D. Multiscale Relation Exploration

Unlike the previous segmentation studies, an MRE module is proposed to further exploit global context information and explore the coupling relation between defect and weld, which consists of multiscale feature aggregation and coupling relation mining.

1) Multiscale Feature Aggregation: As shown in Fig. 3, the enhanced feature from SAM is x. To obtain global contextual information, a multiscale feature aggregation method based on different regions is proposed. The feature aggregation is implemented through three different convolution kernels. More formally, the feature aggregation $G(\cdot)$ can be defined as

$$G(x) \triangleq \alpha \text{conv}(x, s_1) + \beta \text{conv}(x, s_2) + \gamma \text{conv}(x, s_3) \tag{7}$$

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 17,2026 at 03:48:22 UTC from IEEE Xplore. Restrictions apply.

where $\text{conv}(x, s_i)$ denotes the feature generated by the 2-D convolution with kernel $s_i (i = 1, 2, 3)$. α, β, and γ denotes learnable aggregation parameters. The details are described in the following.

First, features at different regions x_1, x_2, and x_3 are fused by an elementwise addition: $x_a = x_1 + x_2 + x_3$, $x_a \in \mathbb{R}^{C\times H\times W}$, and the global average pooling $g(\cdot)$ is performed to obtain a rough global channel representation x_g

$$x_g = g(x_a) = \frac{1}{HW}\sum_{i=1}^{H}\sum_{j=1}^{W} x_a(i, j). \tag{8}$$

Then, the multiscale 1-D convolution is designed to focus on the correlation between adjacent channels in x_g and generate multiscale channel features, which are further aggregated to obtain fine-grained global features x_m

$$\begin{aligned} x_m &= \mathbb{C}\left(x_{m1}, x_{m2}, \ldots, x_{mn}\right) \\ &= \mathbb{C}\left(f_{k1}(x_g), f_{k2}(x_g), \ldots, f_{kn}(x_g)\right) \end{aligned} \tag{9}$$

where n denotes the number of 1-D convolution, f_{ki} denotes the 1-D convolution with the kernel $k_i (i = 1, 2, \ldots, n)$. Experiments show that the model performs optimally when $n = 3$. Then, the learnable aggregation parameters are generated

$$(\alpha, \beta, \gamma) = T\left(\mathbb{C}\left(\text{conv}_{b1}(x_m), \text{conv}_{b2}(x_m), \text{conv}_{b3}(x_m)\right)\right) \tag{10}$$

where $T(\cdot)$ denotes the Softmax function across channels. Finally, the aggregated feature x_{agg} is calculated in

$$x_{\text{agg}} = \alpha x_1 + \beta x_2 + \gamma x_3,\ \alpha + \beta + \gamma = 1. \tag{11}$$

2) Coupling Relation Mining: Most existing methods ignore the contextual dependencies between defect and weld, so that the coupled features are hardly distinguished. Unlike the above studies, a learnable relation matrix is proposed to further explore the coupling relation between defect and weld, so that independent valid features can be acquired. For the ground truth $\mathbb{G}$, the "coupling relation" is clearly known (i.e., which pixels are defect and which pixels are not). Hence, a relation matrix based on the ground truth should be learned to guide the network in mining the coupling between defect and weld. Toward this end, the ground truth is converted to fingerprint map P by downsampling and one-hot encoding as supervision. The details are explained ahead.

First, the ground truth is downsampled as $L \in \mathbb{R}^{1\times(H\times W)}$, and the one-hot encoding is applied to transform L as $K \in \mathbb{R}^{C'\times(H\times W)}$, where C' is the number of categories. Then, the fingerprint map $P \in \mathbb{R}^{N\times N}$ can be obtained: $P = K^\top K$, where $N = H \times W$. Next, the aggregated features $x_{agg} \in \mathbb{R}^{C\times H\times W}$ is converted to $x_n \in \mathbb{R}^{(H\times W)\times H\times W}$, and the relation matrix $M \in \mathbb{R}^{(H\times W)\times(H\times W)}$ can be obtained: $M = x_n^\top x_n$. Finally, the loss function $\mathcal{L}_r$ is calculated between relation matrix M and fingerprint map P in

$$\mathcal{L}_r = -\frac{1}{N^2}\sum_{n=1}^{N^2}\left(P_n \log M_n + (1 - P_n)\log(1 - M_n)\right). \tag{12}$$

Through the relation matrix, the relation between intraclass and interclass can be clarified, so the features belonging to defect or weld can be adaptively selected. The intraclass features y and the interclass features $\widetilde{y}$ can be obtained in (13). Finally, the features x and both aggregated features are concatenated to generate the segmentation features x_{seg}

$$\begin{cases} y = M \otimes x_{\text{agg}} \\ \widetilde{y} = (\mathbb{I} - M) \otimes x_{\text{agg}} \\ x_{seg} = \mathbb{C}(y, \widetilde{y}, F^*) \end{cases} \tag{13}$$

where x_{agg} is reshaped as $x_{\text{agg}} \in \mathbb{R}^{(H\times W)\times C}$, $\otimes$ denotes the matrix product, and $\mathbb{I}$ denotes the matrix with all elements 1.

The advantages of the proposed MRE are listed.

1) The multiscale feature aggregation enables the features of different regions to be selected adaptively, so that the rich global contextual information can be obtained.
2) The relation matrix clarifies the coupling relation between defect and weld, so that the intraclass features can be fully aggregated and the interclass differences between defect and weld can be effectively distinguished for coupled regions.

E. Multiview Collaborative Detection

Most of the existing methods are often based on a single decision. However, it is difficult for a single network to achieve accurate decision-making in complex coupled regions. Unlike the single segmentation network studies, a multiview collaborative mechanism is proposed to achieve complementary and coarse-to-fine decision from three views: 1) positive view (PV), 2) negative view (NV), and 3) fine view (FV), so that accurate detection results can be obtained. The details are described as follows.

1) Positive View: In the PV stage, the network is encouraged to learn the response of the target class (i.e., the pixels are defect). First, the segmentation features x_{seg} are sent to 2-D convolution Conv_{pos} to generate the decision response r. Then, r is upsampled into amplified response $R \in \mathbb{R}^{C'\times H'\times W'}$, where H' and W' are the sizes of the input D. Next, the loss function $\mathcal{L}_p$ is calculated between decision response R and ground truth $\mathbb{G}$ as follows:

$$\mathcal{L}_p = -\frac{1}{N'}\sum_{n=1}^{N'}\left(R_n \log \mathbb{G}_n + (1 - R_n)\log\left(1 - \mathbb{G}_n\right)\right). \tag{14}$$

2) Negative View: In the NV stage, the network is expected to learn responses irrelevant to the target class (i.e., the pixels are not defect). As with the PV, the 2-D convolution Conv_{neg} is applied to obtain the decision response u. Noteworthy, in contrast to the PV, to pay more attention to the nontarget class, the reverse response function (RRF) is designed to invert the u to reversed decision response $\widehat{u}$ in (15). Next, $\widehat{u}$ is upsampled as $U \in \mathbb{R}^{C'\times H'\times W'}$, which is adopted in the calculation of cross-entropy loss $\mathcal{L}_n$

$$\widehat{u}(i, j) = \frac{1}{\text{ReLU}\left(\text{Conv}_{\text{neg}}(i, j)\right) + 0.125} - 4. \tag{15}$$

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 17,2026 at 03:48:22 UTC from IEEE Xplore. Restrictions apply.

3) Fine View: The PVs and NVs focus on the salient regions, which have a weaker response to the coupling region. Therefore, in the FV stage, the network is designed to learn the response in the coupled regions (i.e., are the pixels defect or weld). Define the defect region coupled to the weld as Θ. In the PV, Θ has a strong response for both defect and weld, whereas in the NV, Θ has a stronger negative response for welds, and almost no response for defects. Therefore, to amplify the response of Θ to defect, an FV is designed by integrating the decision response from different views. The details are explained as follows.

First, a reverse affinity matrix ϖ based on the PV is calculated to enhance the response of nondefective regions in the NV. ϖ is calculated in

$$\begin{cases} \varpi = T(r_p) + S(r_p) \\ r_p = \frac{1}{\mathrm{ReLU}(\mathrm{Conv}_{\mathrm{pos}}(i,j))+0.125} - 4 \end{cases} \tag{16}$$

where $T(\cdot)$ and $S(\cdot)$ denote the Softmax and sigmoid, respectively. Then, based on the reverse affinity matrix, the response w of the FV is calculated in (17). Next, w is upsampled as $W \in \mathbb{R}^{C' \times H' \times W'}$ to calculate the cross-entropy loss $\mathcal{L}_f$

$$w = r - \varpi \otimes u. \tag{17}$$

In addition, to guarantee the utilization of shallow features, inspired by Chaiyasarn et al. [22], the auxiliary loss is introduced in the third stage of feature extraction. Finally, the loss function is defined in (18). Noteworthy, the outputs of the FV are the detection results. Algorithm 1 summarizes the training process

$$\mathcal{L} = \mathcal{L}_r + \mathcal{L}_p + \mathcal{L}_n + \mathcal{L}_f + \mathcal{L}_{\mathrm{aux}}. \tag{18}$$

The advantages of the MCD are given as follows.

1) The PVs and NVs are based on an opposing hierarchy to realize complementary defect response, which guarantees the completeness of network decision-making.
2) The FV fully integrates the response from different views, achieving a coarse-to-fine decision transformation, so that the response of the coupled regions to weld defect and the detection accuracy of weld defect can be improved.

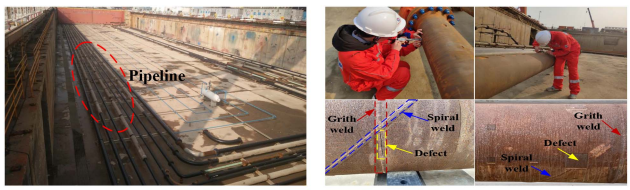
IV. Experiments and Discussions

To confirm the validity of the proposed method, multiple sets of experiments are performed with experimental pipeline signals. Eight advanced methods are adopted to perform a comparative study including: STLR [10] cascaded attention feature fusion network (CAFF-Net) [16], pyramid scene parsing network (PSPNet) [17], criss cross network (CCNet) [18], disentangled nonlocal network (DNLNet) [19], context-based tandem network (CTNet) [20], FCN [22], and DeepLab with multiscale attention network (DMANet) [24]. Among them, STLR is a non-DL method that detects defects in MFL signals by segmentation. CAFF-Net is an advanced region-level detection method that realizes high-accuracy defect detection under complex MFL signals. Other methods are advanced pixel-level DL detection methods. For a fair comparison, the advanced modules of CAFF-Net are applied to the segmented network. Each set of experiments is conducted three times and the mean results are presented.

Algorithm 1: The CRNet Training Process.

Input: Training samples: $\mathbb{D} = \{d_i\}_{i=1}^{N}$, ground truth: $\mathbb{G} = \{g_i\}_{i=1}^{N}$, the number of iteration K.
Output: The trained model of CRNet.

```
Initialize: Parameters of feature extractor, MRE, and MCD: F_a(θ), F_r(ϑ), and F_d(ς).
for iteration = 1 to K do
    Input d_i to F_a(·|θ), extract the sparse features Ψ;
    Input Ψ to the MRE, obtain the relation matrix M and segmentation feature x_seg: M, x_seg = F_r(Ψ|ϑ);
    Calculate the relation loss L_r according to (12);
    Input x_seg to MCD, obtain the multiview predictions: {u, r, w} = F_d(x_seg|ς)
    Apply {u, r, w} and g_i to calculate the multiview loss L_p, L_n, L_f and auxiliary loss L_aux;
    # Update the parameters of CRNet
    Minimize (18) with SGD;
end for
return the trained CRNet model.
```



(a) (b)

Fig. 5. Experimental setup. (a) Pipeline platform. (b) Actual defects.

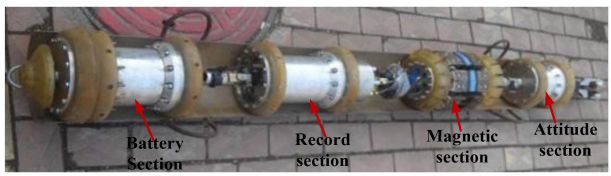


Fig. 6. MFL inspection robot.

A. Experiment Setting

1) Data Acquisition: The experimental data are collected from the pipeline platform in northern China, which is indicated in Fig. 5(a). The pipeline is made of X65 carbon steel. The outside diameter and wall thickness of the measured pipelines are 325 mm and 12.5 mm, respectively. The experimental data are collected by the MFL inspection robot in Fig. 6, which includes an attitude section, magnetic section, record section, and battery section. In operation, the robot's working pressure is 3 MPa, and the running speed range is 0–5 m/s. The Hall sensors are arranged consecutively in the 48 capsules of the magnetic section, where each capsule includes 6 axial Hall sensors (i.e., there are 288 axial Hall sensors in total). The type of the Hall sensors is SS495 A. The sampling interval of the Hall sensors is 0.002 m (i.e., the interval between every two sampling points is 0.002 m). The signal range measured by the sensor is $\pm$ 670 Gauss and the sensor sensitivity is 3.125 $\pm$ 0.125 mV/G.

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 17,2026 at 03:48:22 UTC from IEEE Xplore. Restrictions apply.

TABLE I
MAIN PARAMETER SETTINGS OF THE CRNET

Module	Name	Input	Kernel size, Stride, Padding	Output
SAM	Conv_Q	F	1×1 512 Conv2d + BN, s=1, p=0	x
	Conv_K		1×1 512 Conv2d + BN, s=1, p=0	
	Conv_V		1×1 2048 Conv2d + BN, s=1, p=0	
			Reshape + Multiplication + Addition	
MRE	Conv_{s1}	x	1×1 512 Conv2d + BN, s=1, p=0	x_g
	Conv_{s2}		3×3 512 Conv2d + BN, s=1, p=1	
	Conv_{s3}		[3×3 512 Conv2d + BN, s=1, p=1]×2	
			Addition + Global average pooling	
	k_1	x_g	7×1 512 Conv1d + BN + ReLU, s=1, p=3	x_m
	k_2		13×1 512 Conv1d + BN + ReLU, s=1, p=6	
	k_3		25×1 512 Conv1d + BN + ReLU, s=1, p=12	
			Addition	
	Conv_{b1}	x_m	1×1 512 Conv1d + BN + ReLU, s=1, p=0	x_{agg}
	Conv_{b2}		1×1 512 Conv1d + BN + ReLU, s=1, p=0	
	Conv_{b3}		1×1 512 Conv1d + BN + ReLU, s=1, p=0	
			Concat + Softmax + Multiplication + Addition	
	Conv_{z1}	x_{agg}	1×1 2106 Conv2d + BN, s=1, p=0	x_{seg}
			Reshape + Transpose + Multiplication + Concat	
	Conv_{z2}		1×1 512 Conv2d + BN, s=1, p=0	
MCD	Conv_{reg}	x_{seg}	1×1 4 Conv2d + BN, s=1, p=0	u
	Conv_{pos}		1×1 4 Conv2d + BN, s=1, p=0	r
			Sigmod + Softmax + Multiplication + Subtraction	w

[1] "BN" denotes "Batch Normalization".

2) MFL Dataset Description: In the experiments, 100-m signals (containing $288 \times (100 \div 0.002) = 288 \times 50\,000$ sampling points) are applied. To ensure the signals' integrity, the raw signals are interpolated, where the number of sampling points changes from $288 \times 50\,000$ to $308 \times 50\,000$. The training dataset is constructed as follows: First, the raw signals are segmented based on a sliding window of 308×432 and a fixed shift step of 50. Then, to highlight the features of the defect, the segmented signals are converted to training images based on the proposed signal transformation. Finally, the invalid signals are removed and the standard training dataset is generated. The dataset contains 841 samples including Normal, spiral weld (Spiral), girth weld (Girth), and weld defect (Defect), which are shown in Fig. 5(b). The dataset is randomly divided into a training set and a testing set based on 8:2, where there is no overlap between them.

3) Network Setting and Evaluation Metrics: The backbone of CRNet is based on pretrained ResNet-50, where the last two stages of downsampling are removed. The main parameter settings of the CRNet are illustrated in Table I. The network's optimizer and batch size are stochastic gradient descent (SGD) and 8, respectively. The initial learning rate is 0.001, where its optimization method is MultiStepLR. The number of iterations is 30 epochs. The size of the training sample is 308×432. All codes are realized under the Pytorch framework. The configuration of the computer is listed ahead, Operating System: Ubuntu 18. 04. 5 LTS, CPU: 2 AMD EPYC 7601 2.7 GHz 32-Core Processor, Memory: 128 GB, GPU: 2 NVIDIA GeForce RTX 3090, and Hard disc: 1 TB.

Similar to the work in [28], to evaluate the detection accuracy of each category, the pixelwise accuracy (true positive rate) is defined as $Accuracy$, which can reflect the model's missed diagnosis. The false positive rate (FPR) is applied to reflect the model's misdiagnosis. The $F1-\text{score}(F1)$ is applied as a composite measure of the model's missed diagnosis and misdiagnosis. In addition, the intersection over union (IoU) is applied to evaluate the location accuracy. Noteworthy, in the MFL detection system, the missed diagnosis is of more concern to us than the misdiagnosis because the missed diagnosis of weld defects will result in serious accidents. The evaluation metrics are defined as follows:

$$\text{Accuracy} = \frac{\text{TP}}{\text{TP}+\text{FN}},\ \text{IoU} = \frac{O_p \cap O_g}{O_p \cup O_g} \tag{19}$$

$$\text{FPR} = \frac{\text{FP}}{\text{FP}+\text{TN}},\ F1 = \frac{2 \times \text{TP}}{2 \times \text{TP}+\text{FP}+\text{FN}} \tag{20}$$

where TP, FP, TN, and FN denote the true positive, false positive, true negative, and false negative, respectively. O_p and O_g denote the prediction set and label set, respectively. For a certain category, $O_p \cap O_g$ denotes that both the prediction and the label are the set of pixels of this category. $O_p \cup O_g$ denotes that at least one of the predictions and labels is a set of pixels of this category.

B. Experiment 1: Comparison of Accuracy

The experiments are conducted to demonstrate the validity of the proposed method for the detection of weld defect. The experimental results are illustrated in Table II. The experimental results can be concluded that the overall performance of the proposed method is superior to other comparative methods. For non-DL methods (STLR), the effective features of weld defect are hardly selected, and the complex coupling relation reduces the performance of defect detection. For DL methods, in terms of $Accuracy$, the results are inferior to the proposed method, whose average detection accuracy can reach 95.5%. In particular, the defect detection accuracy reaches 91.6%, which is on average 11.2% higher than other methods. The results show that our method is less prone to missed diagnosis during defect detection, which is essential to ensure the safe operation of the pipeline. In addition, in terms of the FPR, the proposed method can reach 1.52%, which significantly outperforms other comparison methods. The results show that the proposed method has a low misdiagnosis rate, which guarantees the accuracy of the defect detection. In terms of the $F1$, the proposed method can reach 95.5% on average, which better than other methods. The results prove that the proposed method outperforms other methods in terms of both overall detection accuracy and misdiagnosis. In terms of IoU, the proposed method can reach 87.8%, which is on average 5.0% higher than other methods. This indicates that our method has more accurate location accuracy. Fig. 7 illustrates the results of weld defect detection. In general, the proposed method has better detection accuracy and location accuracy, and the experimental results demonstrate the validity of the proposed method in the weld defect detection.

The reasons for the excellent detection performance are summarized as follows: 1) Compared with other methods, the proposed method can focus more on valid weld and defect information. In addition, weld features and defect features in

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 17,2026 at 03:48:22 UTC from IEEE Xplore. Restrictions apply.

TABLE II
RESULTS OF DIFFERENT COMPARISON METHODS IN EXPERIMENT 1

Methods	Accuracy					IoU					FPR					$F1$				
	Normal	Spiral	Girth	Defect	Mean.	Normal	Spiral	Girth	Defect	Mean.	Normal	Spiral	Girth	Defect	Mean.	Normal	Spiral	Girth	Defect	Mean.
STLR [10]	98.5%	82.9%	70.9%	63.9%	79.1%	96.2%	77.0%	68.2%	58.6%	75.0%	25.3%	0.42%	1.97%	0.28%	7.00%	71.8%	90.0%	80.2%	77.6%	79.9%
CAFF-Net [16]	99.4%	87.9%	84.5%	80.3%	88.0%	98.6%	82.1%	77.6%	72.0%	82.6%	13.5%	0.32%	1.90%	0.15%	3.97%	82.8%	93.1%	88.9%	88.9%	88.4%
PSPNet [16]	99.1%	90.2%	77.7%	78.9%	86.5%	98.2%	83.4%	76.4%	73.2%	82.8%	16.3%	0.43%	1.13%	0.16%	4.50%	80.0%	94.2%	85.8%	88.0%	87.0%
CCNet [18]	99.3%	89.5%	89.8%	81.0%	89.9%	98.1%	82.9%	84.2%	74.5%	84.9%	11.6%	0.40%	1.24%	0.18%	3.36%	84.9%	93.9%	92.8%	89.2%	90.2%
DNLNet [19]	99.4%	90.5%	91.6%	82.6%	91.0%	98.4%	83.1%	84.6%	74.0%	85.0%	9.3%	0.30%	2.22%	**0.14%**	3.00%	87.4%	94.6%	92.4%	90.3%	91.2%
CTNet [20]	99.4%	88.8%	90.7%	82.0%	90.2%	98.5%	82.0%	83.9%	72.0%	84.1%	10.9%	**0.22%**	1.70%	0.25%	3.27%	85.6%	93.7%	92.6%	89.7%	90.4%
FCN [22]	99.3%	89.6%	90.8%	76.8%	89.1%	98.5%	82.8%	81.4%	71.0%	83.4%	12.1%	0.36%	1.92%	0.17%	3.63%	84.3%	94.0%	92.4%	86.7%	89.4%
DMANet [24]	99.2%	88.9%	92.3%	81.4%	90.4%	98.4%	82.5%	85.7%	73.0%	84.9%	10.5%	0.24%	1.84%	0.18%	3.20%	86.1%	93.7%	93.3%	89.4%	90.6%
Proposed	**99.6%**	**93.6%**	**97.0%**	**91.6%**	**95.5%**	**98.8%**	**84.4%**	**87.7%**	**80.2%**	**87.8%**	**5.1%**	0.34%	**0.50%**	0.15%	**1.52%**	**92.7%**	**96.2%**	**97.7%**	**95.4%**	**95.5%**

The boldface values are used to highlight optimal experimental results.

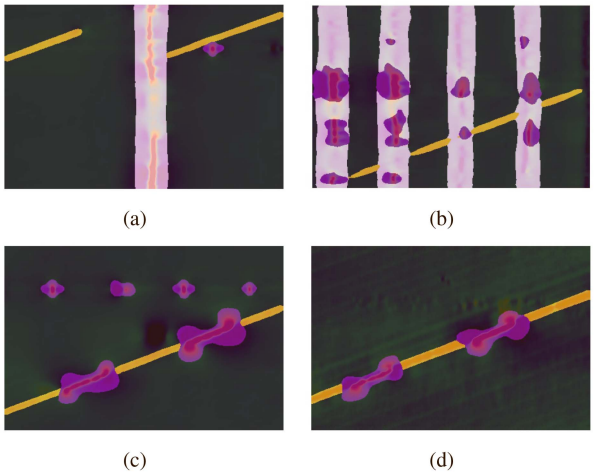


Fig. 7. Examples of detection results. Different colors denote different classes. Purple, white, and yellow denote defects, girth welds, and spiral welds, respectively. Others denote normal signals. (a) Denotes the detection results of the girth weld and spiral weld. (b) Denotes the detection results of the defect in girth weld. (c) and (d) Denote the detection results of the defect in spiral weld.

the coupling regions are effectively distinguished by the proposed MRE, which provides the basis for highly accurate weld defect detection. 2) The proposed MCD mechanism realizes the complementary and coarse-to-fine decision-making judgment of coupled regions, so that more accurate detection results and location results can be obtained.

C. Experiment 2: Comparison of Robustness

The MFL inspection robot often works in complex noise environments, the acquired signal usually contains noise. To demonstrate the robustness of the proposed method, multiple noise experiments are conducted with the same setup as Experiment 1. In addition, to evaluate the overall performance of the method, $Accuracy$ and IoU are applied as evaluation metrics. In the process, the Gaussian white noise with different signal-to-noise ratios (SNR) is mixed into the original signal. The SNR is defined in (21). Specifically, the SNR of the original signal is 50 dB. The SNR of the synthesized noise signals are 45 dB, 40 dB, 35 dB, and 30 dB, respectively

$$\mathrm{SNR} = 10\log_{10}\frac{P_{\mathrm{signal}}}{P_{\mathrm{noise}}} \tag{21}$$

where P_{signal} and P_{noise} denote the power of the original signal and noise, respectively. The experimental results are presented in Fig. 8. It can be found that as the SNR decreases, the $Accuracy$ and IoU of all methods decreases but the performance of the proposed method remains optimal. In terms of $Accuracy$, the defect detection accuracy is still above 80% when the SNR is higher than 35 dB. In addition, when the SNR is 30 dB, the girth weld detection accuracy of the proposed method is still above 90%, and the detection accuracy of defect and weld is on average 14.3% higher than comparative methods. In terms of IoU, as the SNR decreases, the IoU of the proposed method decreases by only 6.4%, which is significantly better than other methods, which indicates that the proposed method possesses better segmentation boundaries even under noise interference. Overall, the proposed method outperforms other methods in weld defect detection accuracy and location accuracy under different noise interference.

The reasons for the stronger robustness can be summarized as follows: 1) Compared to non-DL methods, the proposed method can obtain more effective deep features and reduce the reliance on shallow features, where noise signals affect more shallow features such as amplitude, texture, etc. 2) Compared with other DL methods, the proposed method can predict the results from multiple views and reduce the reliance on a single decision, so that the accuracy of decisions under noise interference can be guaranteed.

D. Experiment 3: Comparison of Model Complexity

To verify the model complexity of the proposed method, the parameter quantity and floating point operations (FLOPs) for different DL comparison methods are calculated in the MMSegmentation framework. Table III illustrates the comparison results. The results indicate that the proposed method is not optimal in terms of parameter quantity and FLOPs, but the accuracy of defect detection is significantly improved. Overall, the computational complexity of the proposed method is at a medium stage. Noteworthy, the data analysis process in the MFL

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 17,2026 at 03:48:22 UTC from IEEE Xplore. Restrictions apply.

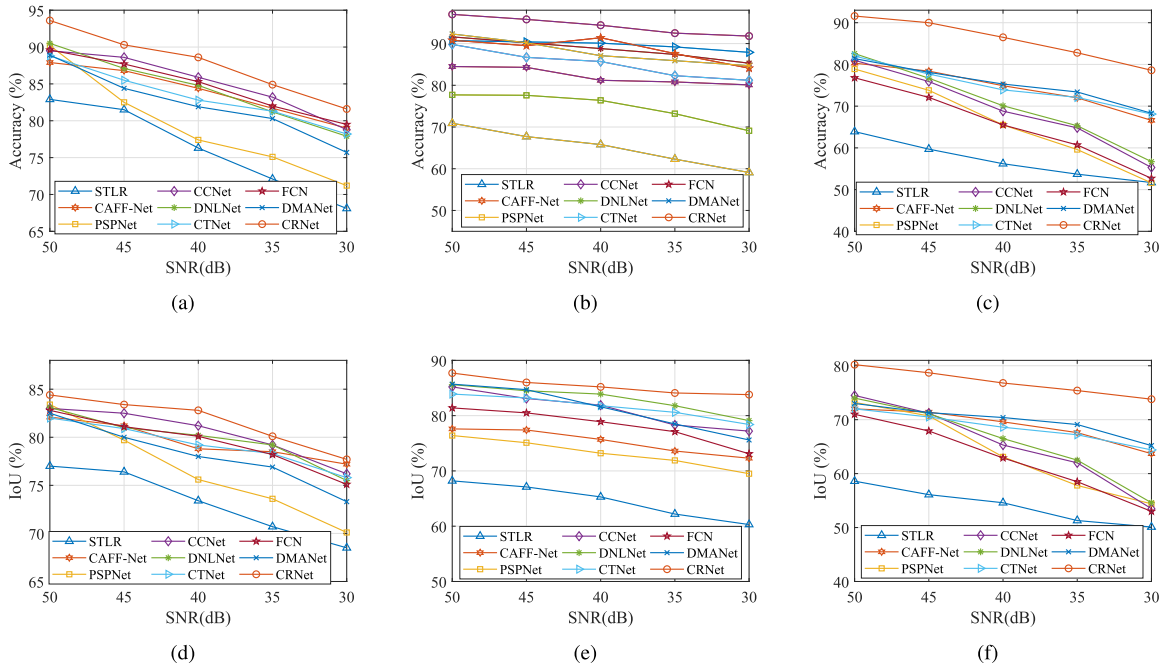


Fig. 8. Experimental results at different signal-to-noise ratios. (a), (b), and (c) denote the $Accuracy$ of spiral weld, girth weld, and defect, respectively. (d), (e), and (f) denote the IoU of spiral weld, girth weld, and defect, respectively.

TABLE III
COMPARISON OF PARAMETER QUANTITY AND FLOPS OF DIFFERENT METHODS

Methods	Parameters (M)	GFLOPs
CAFF-Net [16]	76.98	147.15
PSPNet [17]	48.96	91.74
CCNet [18]	50.34	102.99
DNLNet [19]	50.01	102.71
CTNet [20]	63.65	115.06
FCN [22]	49.49	101.61
DMANet [24]	67.59	119.52
Proposed	70.13	133.17

TABLE IV
RESULTS OF ABLATION EXPERIMENTS ON SAM

Model	Accuracy			IoU		
	Spiral	Girth	Defect	Spiral	Girth	Defect
w/o SAM	90.8%	93.6%	84.1%	82.0%	86.1%	73.6%
S4+SAM	92.6%	95.1%	88.0%	83.9%	87.1%	79.5%
S3+S4+NL [27]	92.4%	95.2%	86.2%	82.6%	86.3%	75.8%
S3+S4+SAM	**93.6%**	**97.0%**	**91.6%**	**84.4%**	**87.7%**	**80.2%**

[1] "w/o" denotes "without".
[2] S3 and S4 denote the third and fourth stages of feature extraction network, respectively.
The boldface values are used to highlight optimal experimental results.

detection system is an offline process, and model complexity does not directly affect the performance of weld defect detection. Therefore, the proposed method has good capability in MFL weld defect detection.

E. Experiment 4: Ablation Experiments

The performance of the proposed method is mainly related to three core components including SAM, MRE module, and MCD. To demonstrate the effectiveness of each core component, multigroup ablation studies are carried out. The detailed analysis is as follows.

1) Analysis of SAM: In the proposed method, SAM is designed to enhance the ability of the network to focus on core areas. To verify the validity of SAM, it is compared to the nonlocal neural networks (NL) [27], which is proved to be effective in many tasks. In addition, the sensitivity analysis is performed for the location of SAM in the feature extraction network. The experimental results are shown in Table IV. It can be concluded that SAM is more effective than NL. When SAM is applied, the detection and location accuracy of the model are improved overall. In particular, the detection accuracy can be improved by 4.6% on average. In addition, compared with NL, the overall performance of the proposed method is increased, especially the detection and location accuracy of defect can be improved by 5.4% and 4.4%, respectively. From Fig. 9, it can be seen that the SAM enables the network to pay more attention to effective features compared to NL networks. Meanwhile, the performance of the model is optimized when SAM is applied to S3 and S4. This is mainly because the attention mechanism can better aggregate the deep abstract features of both S3 and S4, and the continuous joint optimization makes the model extract more effective features, so the performance is optimal.

2) Analysis of MRE: The MRE is proposed to mine the coupling relation between defect and weld, which is important for weld defect detection. We verify the importance of MRE and the effect of the core parameter k on the model. The

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 17,2026 at 03:48:22 UTC from IEEE Xplore. Restrictions apply.

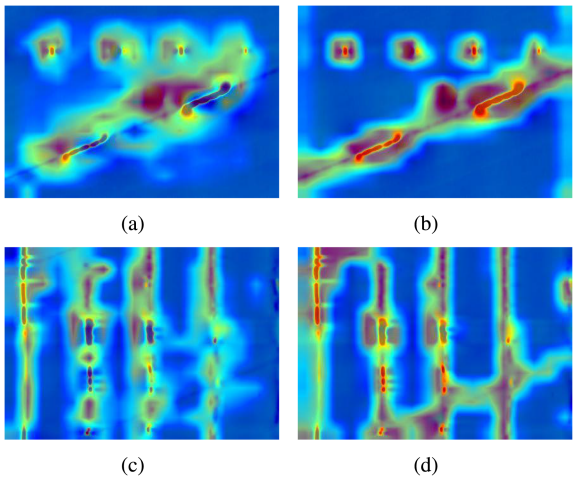


Fig. 9. Examples of attention heatmaps. (a) and (c) Attention heatmaps of NL. (b) and (d) Attention heatmaps of SAM.

TABLE V
RESULTS OF ABLATION EXPERIMENTS ON MRE

Model	Accuracy			IoU		
	Spiral	Girth	Defect	Spiral	Girth	Defect
w/o MRE	91.1%	90.0%	80.1%	82.7%	80.4%	73.2%
k={7}	91.3%	95.2%	82.6%	83.5%	86.1%	76.9%
k={7, 13}	91.7%	94.4%	85.3%	83.8%	87.4%	77.5%
k={7, 13, 25}	**93.6%**	**97.0%**	**91.6%**	**84.4%**	**87.7%**	**80.2%**
k={7, 13, 25, 49}	90.5%	94.6%	88.2%	83.6%	87.6%	79.5%

The boldface values are used to highlight optimal experimental results.

experimental results are shown in Table V. It can be observed that when the MRE is not applied, the detection accuracy and location accuracy of the model decreased by 7.0% and 5.3%, respectively. Compared with single-scale ($k = 7$), multiscale can better exploit global contextual information and improve the detection accuracy. However, when $k = \{7, 13, 25, 49\}$, the detection accuracy and location accuracy of the proposed method are reduced. This is mainly because more remote channel information with low correlation is fused as the multiscale increases, which increases redundant information, so that the performance of the model is reduced.

3) Analysis of MCD: The MCD is proposed to realize weld defect detection from multiple views. In the decision-making process, the three views reinforce each other. To verify the necessity of the various branches of MCD including PV, NV, and FV, a series of ablation studies are conducted. In the decision-making process, the decision network for each view is filtered separately. The experimental results are shown in Table VI, where the performance is degraded by the absence of any one view. Specifically, when the FV is not applied, the detection and location accuracy are reduced by 4.4% and 3.4% on average, respectively. This experiment demonstrates the importance of an MCD mechanism in the weld defect detection. In general, the proposed MCD mechanism achieves complementary decision-making and ensures more accurate detection results.

TABLE VI
RESULTS OF ABLATION EXPERIMENTS ON MCD

Model	Accuracy			IoU		
	Spiral	Girth	Defect	Spiral	Girth	Defect
w/o PV	90.4%	95.6%	88.1%	83.5%	85.6%	77.1%
w/o NV	91.8%	94.9%	88.3%	84.0%	84.8%	77.6%
w/o FV	88.3%	94.1%	86.5%	82.7%	83.7%	75.8%
Proposed	**93.6%**	**97.0%**	**91.6%**	**84.4%**	**87.7%**	**80.2%**

The boldface values are used to highlight optimal experimental results.

V. CONCLUSION

This article proposed a novel MFL weld defect detection method, which has achieved high-accuracy weld defect detection. First, the proposed feature extraction network with sparse attention reduced redundant weld features coupled in defects, so that effective weld defect-related sparse features can be acquired. Second, the proposed MRE module fully exploited the coupling relation between defect and weld, so that high-accuracy weld defect detection can be guaranteed. Finally, the proposed MCD mechanism enabled complementary and coarse-to-fine decision-making, so that more accurate detection results can be obtained. The proposed method realized pixel-level MFL weld defect detection for the first time and clarified the coupling relation between defect and weld. The experimental results have shown that the proposed method is effective in MFL weld defect detection, which indicates its promise in engineering.

However, there are some limitations in the proposed method. The proposed method has a slightly higher number of parameters and computational complexity than other high-accuracy comparison methods. This limits the application of the proposed method in specific industrial scenarios. Therefore, in future work, we will conduct in-depth research on high-accuracy lightweight models.

REFERENCES

[1] X. Hu, H. Zhang, D. Ma, R. Wang, and P. Tu, "Small leak location for intelligent pipeline system via action-dependent heuristic dynamic programming," *IEEE Trans. Ind. Electron.*, vol. 69, no. 11, pp. 11723–11732, Nov. 2022.

[2] L. Wang, H. Zhang, J. Liu, F. Qu, and F. Zuo, "Defect size quantification for pipeline magnetic flux leakage detection system via multi-level knowledge-guided neural network," *IEEE Trans. Ind. Electron.*, vol. 70, no. 9, pp. 9550–9560, Sep. 2023.

[3] Y. Shi, C. Shen, H. Fang, and H. Li, "Advanced control in marine mechatronic systems: A survey," *IEEE/ASME Trans. Mechatron.*, vol. 22, no. 3, pp. 1121–1131, Jun. 2017.

[4] L. Jiang, H. Zhang, J. Liu, X. Shen, and H. Xu, "A multisensor cycle-supervised convolutional neural network for anomaly detection on magnetic flux leakage signals," *IEEE Trans. Ind. Informat.*, vol. 18, no. 11, pp. 7619–7627, Nov. 2022.

[5] M. Fu, J. Liu, H. Zhang, and S. Lu, "Multisensor fusion for magnetic flux leakage defect characterization under information incompletion," *IEEE Trans. Ind. Electron.*, vol. 68, no. 5, pp. 4382–4392, May 2021.

[6] Z. Zhou and Z. Liu, "Fault diagnosis of steel wire ropes based on magnetic flux leakage imaging under strong shaking and strand noises," *IEEE Trans. Ind. Electron.*, vol. 68, no. 3, pp. 2543–2553, Mar. 2021.

[7] X. Yang, Y. Zhou, and H. Liu, "Wearable ultrasound-based decoding of simultaneous wrist/hand kinematics," *IEEE Trans. Ind. Electron.*, vol. 68, no. 9, pp. 8667–8675, Sep. 2021.

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 17,2026 at 03:48:22 UTC from IEEE Xplore. Restrictions apply.

[8] H. M. Kim and D.-H. Choi, "Defects detection of gas pipeline near the welds based on self quotient image and discrete cosine transform," *Russian J. Nondestruct. Testing*, vol. 52, no. 3, pp. 175–183, 2016.
[9] X. Gao, X. Zhou, C. Wang, N. Ma, Y. Zhang, and D. You, "Skin depth and detection ability of magneto-optical imaging for weld defects in alternating magnetic field," *J. Manuf. Syst.*, vol. 55, pp. 44–55, 2020.
[10] M. Fu, J. Liu, D. Zang, and S. Lu, "Anomaly detection of complex MFL measurements using low-rank recovery in pipeline transportation inspection," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 9, pp. 6776–6786, Sep. 2020.
[11] H. Jiang, Y. Zhao, J. Gao, and Z. Wang, "Weld defect classification based on texture features and principal component analysis," *Insight-Non-Destructive Testing Condition Monit.*, vol. 58, no. 4, pp. 194–200, 2016.
[12] X. Shen, J. Liu, J. Sun, L. Jiang, H. Zhao, and H. Zhang, "SSCT-Net: A semisupervised circular teacher network for defect detection with limited labeled multiview MFL samples," *IEEE Trans. Ind. Informat.*, to be published, doi: 10.1109/TII.2022.3232764.
[13] Y. Wang, Y. Lei, N. Li, T. Yan, and X. Si, "Deep multisource parallel bilinear-fusion network for remaining useful life prediction of machinery," *Rel. Eng. Syst. Saf.*, vol. 231, 2023, Art. no. 109006.
[14] T. Feng, S. Huang, J. Liu, J. Wang, and X. Fang, "Welding surface inspection of armatures via CNN and image comparison," *IEEE Sensors J.*, vol. 21, no. 19, pp. 21696–21704, Oct. 2021.
[15] X. Lang, "Recognition method of pipeline weld defects based on auxiliary classifier generative adversarial networks," *IEEE Instrum. Meas. Mag.*, vol. 25, no. 2, pp. 69–77, Feb. 2022.
[16] J. Liu, X. Shen, J. Wang, L. Jiang, and H. Zhang, "An intelligent defect detection approach based on cascade attention network under complex magnetic flux leakage signals," *IEEE Trans. Ind. Electron.*, vol. 70, no. 7, pp. 7417–7427, Jul. 2023.
[17] X. Long, W. Zhang, and B. Zhao, "PSPNet-SLAM: A semantic SLAM detect dynamic object by pyramid scene parsing network," *IEEE Access*, vol. 8, pp. 214685–214695, 2020.
[18] Z. Huang et al., "CCNet: Criss-cross attention for semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 6, pp. 6896–6908, Jun. 2023, doi: 10.1109/ICCV.2019.00069.
[19] M. Yin et al., "Disentangled non-local neural networks," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 191–207.
[20] Z. Li, Y. Sun, L. Zhang, and J. Tang, "CTNet: Context-based tandem network for semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 12, pp. 9904–9917, Dec. 2022.
[21] Y. Chang and W. Wang, "A deep learning-based weld defect classification method using radiographic images with a cylindrical projection," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–11, 2021, doi: 10.1109/TIM.2021.3124053.
[22] K. Chaiyasarn, A. Buatik, H. Mohamad, M. Zhou, S. Kongsilp, and N. Poovarodom, "Integrated pixel-level CNN-FCN crack detection via photogrammetric 3D texture mapping of concrete structures," *Automat. Construction*, vol. 140, 2022, Art. no. 104388.
[23] L. Yang, S. Song, J. Fan, B. Huo, E. Li, and Y. Liu, "An automatic deep segmentation network for pixel-level welding defect detection," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–10, 2022, doi: 10.1109/TIM.2021.3127645.
[24] X. Sun, Y. Xie, L. Jiang, Y. Cao, and B. Liu, "DMA-Net: Deeplab with multi-scale attention for pavement crack segmentation," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 10, pp. 18392–18403, Oct. 2022.
[25] B. Wang, Y. Lei, N. Li, and W. Wang, "Multiscale convolutional attention network for predicting remaining useful life of machinery," *IEEE Trans. Ind. Electron.*, vol. 68, no. 8, pp. 7496–7504, Aug. 2021.
[26] R. Chen, D. Cai, X. Hu, Z. Zhan, and S. Wang, "Defect detection method of aluminum profile surface using deep self-attention mechanism under hybrid noise conditions," *IEEE Trans. Instrum. Meas.*, vol. 70, 2021, Art. no. 3524509, doi: 10.1109/TIM.2021.3109723.
[27] X. Wang, R. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proc. IEEE Conf. Comput. Vis. pattern Recognit.*, 2018, pp. 7794–7803.
[28] X. Zhang, Z. Chen, Q. M. J. Wu, L. Cai, D. Lu, and X. Li, "Fast semantic segmentation for scene perception," *IEEE Trans. Ind. Informat.*, vol. 15, no. 2, pp. 1183–1192, Feb. 2019.



Xiangkai Shen (Graduate Student Member, IEEE) received the B.S. degree in automation from the Shandong University of Science and Technology, Qingdao, China, in 2018, and the M.S. degree in control engineering in 2020 from Northeastern University, Shenyang, China, where he is currently working toward the Ph.D. degree in control science and engineering.

His research interests include fault diagnosis, industrial Big Data analysis, and deep learning.



Jinhai Liu (Senior Member, IEEE) received the B.S. degree in automation from the Harbin Institute of Technology, Harbin, China, in 2002, and the M.S. degree in power electronics and power transmission and the Ph.D. degree in control theory and control engineering from Northeastern University, Shenyang, China, in 2005 and 2009, respectively.

He is currently a Professor and the Doctoral Supervisor with Northeastern University. His research interests include data-driven fault diagnosis, industrial Big Data analysis, and safety technology of long pipelines.



Lin Jiang (Member, IEEE) received the B.S. and M.S. degrees in automation and the degree in control engineering in 2016 and 2018, respectively, from Northeastern University, Shenyang, China, where she is currently working toward the Ph.D. degree in control theory and control engineering with the School of Information Science and Engineering.

Her research interests include nondestructive testing, magnetic flux leakage preprocessing, signal processing, and machine learning.



Xiaoyuan Liu (Student Member, IEEE) received the B.S. degree in electronic information science and technology from Shanxi Normal University, Linfen, China, in 2018, and the M.S. degree in power system and automation in 2020 from Northeastern University, Shenyang, China, where she is currently working toward the Ph.D. degree in control science and engineering with the School of Information Science and Engineering.

Her current research interests include fault diagnosis, nondestructive testing, and deep learning.



Huaguang Zhang (Fellow, IEEE) received the B.S. and M.S. degrees in control engineering from the Northeast Dianli University of China, Jilin City, China, in 1982 and 1985, respectively, and the Ph.D. degree in thermal power engineering and automation from Southeast University, Nanjing, China, in 1991.

In 1992, he joined, as a Postdoctoral Fellow for two years, the Department of Automatic Control, Northeastern University, Shenyang, China, where he has been a Professor and the Head of the Institute of Electric Automation, School of Information Science and Engineering, since 1994. He has authored or coauthored more than 280 journal articles and conference papers and 6 monographs and coinvented 90 patents. His research interests include fuzzy control, stochastic system control, neural-network-based control, nonlinear control, and their applications.

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 17,2026 at 03:48:22 UTC from IEEE Xplore. Restrictions apply.